INCH-POUND
MIL-PRF-53024C(AT)
09 May 2012
SUPERSEDING
MIL-C-53024B(ME)
30 October 1992

PERFORMANCE SPECIFICATION

COMPACTOR, HIGH SPEED, SELF-PROPELLED, COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

Inactive for new design after 09 March 1998.

This specification is approved for use within the TACOM LCMC, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high speed, self-propelled, diesel powered, tamping foot, embankment compactor.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 3895

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2104	-	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-62122	-	Cable Assembly, Inter-Vehicle Power: Plug, Receptacle, And Adapter
MIL-T-704	-	Treatment and Painting of Material.
MIL-PRF-46176	-	Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-V-81940/1	-	Valve, Sampling and Bleed, Hydraulic, Type II Systems
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MS52149	-	Battery, Storage, Lead Acid (Low Maintenance).

FEDERAL STANDARDS

FED-STD-595/34094	-	Green 383

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-209	-	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-464	-	Electromagnetic Environmental Effects for Systems
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1474	-	Noise Limits for Army Materiel.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50271	-	Plate, Identification
A-A-52557	-	Fuel Oil, Diesel for Posts, Camps and Stations.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-113	-	Guide for The Selection of Lubricants, Functional Fluids, Preservatives, and Specialty Products for Use in Ground Equipment Systems.
MIL-HDBK-1791	-	Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

(Copies of these documents are available from https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

US ARMY TANK-AUTOMOTIVE COMMAND (TACOM)

Report No. CR-82-588-003	-	STE/ICE Design Guide for Vehicle Diagnostic Connector Assembly.

US ARMY DRAWINGS

12258941	-	Connector Receptacle 54 PIN.
12258955	-	Vehicle Test Card.
13229E8014	-	Bracket Assembly Rifle Stowage.

(Copies of these documents are available from usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM D1557	-	Moisture-Density Relations of Soils and Soil-Aggregate Mixtures - Unified Soils Classification System using 10-lb. (4.54-KG) Rammer and 18-in. (457-MM) Drop.
ASTM D2487	-	Classification of Soils for Engineering Purposes - AASHO.
ASTM D6938	-	Density of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth).

(Copies of these documents are available from www.astm.org or ASTM International, P.O. Box C700, West Conshohocken, PA 19428-2959.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

INTERNATIONAL STANDARDS ORGANIZATION

ISO 3411	-	Earth-moving Machinery – Physical dimensions of operators and minimum operator space envelope.
ISO 3471	-	Earth-moving machinery — Roll-over protective structures — Laboratory tests and performance requirements.
ISO 6682	-	Earth-moving Machinery – Zones of Comfort and Reach for Controls.
ISO 10570	-	Steering Frame Lock for Articulated Loaders and Tractors.

(Copies of these documents are available from www.iso.org or www.ansi.org or ANSI Customer Service Department, 25 W. 43rd Street, 4th Floor, New York, NY 10036.)

SAE INTERNATIONAL

SAE J88	-	Sound Measurement-Earthmoving Machinery-Exterior
SAE J185	-	Access Systems for Off-Road Machines.
SAE J297	-	Operator Controls on Industrial Equipment.
SAE J386	-	Operators Restraint Systems for Off-Road Work Machines
SAE J514	-	Hydraulic Tube Fittings.
SAE J534	-	Lubrication Fittings.
SAE J537	-	Storage Batteries.
SAE J753	-	Lubrication Chart - Construction and Industrial Machinery.
SAE J899	-	Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines.
SAE J994	-	Alarm - Backup - Electric - Performance Test and Application.
SAE J1362	-	Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines.
SAE J1472	-	Braking Performance - Rollers
SAE J2360	-	Lubricating Oil, Gear Multipurpose (Metric) Use.

(Copies of these documents are available from www.sae.org or SAE Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The compactor shall be a self-propelled, diesel powered, tamping machine for high speed embankment compaction. The compactor shall be a commercial model consisting of articulated steering, a hydraulically controlled strike-off dozer blade, frame or wheel oscillation, and all wheels shall have tamping foot or segmented impact pads with adjustable wheel cleaners (see 3.1.1). The compactor shall be complete with all components furnished as standard equipment that

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

are customarily offered with a compactor of this type whether stipulated herein or not, together with such attachments as may be necessary to enable the compactor to function reliably and efficiently in a sustained operation. The compactor shall conform to all Environmental Protection Agency (EPA) and Occupational Safety and Health Administration (OSHA) legal requirements for US domestic equipment in effect at the date of manufacture.

3.1.1 Commerciality. The compactor shall be the offeror's standard commercial products, as described by the offeror's published commercial literature, except as modified to meet the requirements of this specification, which has been manufactured and sold to the general public in substantial quantities for at least one year prior to the opening date of the Governments solicitation (see 6.4).

3.1.2 Dimensions. The compactor shall not exceed the following dimensions:

Length - 236 inches (without blade.)
Width - 121 inches (without blade.)
Height - 115 inches (with vehicle in transport configuration)

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Performance. The compactor shall be capable of compacting two different groups of soils to at least 1500 cubic yards per hour, each, in each of the two different types of soils cited herein. The two types of soils shall correspond to a clayey sandy (sc) type soil with a plasticity index of 5-12, and a lean clay (cl) type soil with a plasticity index of 10-16. Both types of soils, when compacted, shall be within 2 percentage points of their respective optimum moisture content. In addition to compacting the types of soils cited herein to at least 1500 cubic yards per hour, the compacted soils must be compacted to a density of 90 percent of ASTM D1557 for the entire depth of the lift minus 1 inch maximum. The minimum height of the lift shall be 6 inches. The compactor shall be capable of light bulldozing and leveling of loose material. The compactor shall be capable of climbing a 30 degree slope and shall be capable of turning and maneuvering on a 15 degree slope. The compactor shall be capable of starting and operating as specified herein.

3.4 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials, coatings and special treatments utilized shall provide protection against the various forms of material deterioration in the applicable operating and storage environments to which the compactor may be exposed.

3.4.1 Material deterioration prevention and control. The compactor shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the compactor may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

3.4.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the compactor may be newly fabricated from recovered materials to the maximum extent practicable, provided the compactor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the compactor.

3.4.4 Radioactive material. No radioactive material shall be used in any part of the compactor.

3.4.5 Asbestos. No asbestos material shall be used in any form or in any part of the compactor.

3.5 Maintenance.

3.5.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement. Covers or plates which must be removed or opened for component adjustment or maintenance shall be equipped with quick-opening, hand-operable fasteners. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for hand tools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using common tools. After completion of any maintenance operation, the compactor shall operate as specified herein.

3.5.2 Safety. All exposed parts which are subject to high operating temperatures, or which are energized electrically shall be insulated, fully enclosed or guarded. All rotating, reciprocating or moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. All walking surfaces/steps shall be provided with an anti-skid surface. Seat belts shall be anchored in accordance with SAE J386. The operator's vision shall not be obstructed and shall permit operation in forward and reverse travel. All platforms, steps, and hand holds shall be in accordance with SAE J185 and permit safe entry and exit from the equipment. A reverse alarm of the audible type shall be provided conforming to SAE J994, type C. The backup alarm signal shall be audible above and distinguishable from the surrounding compactor noise level.

3.5.2.1 Rollover protective structure (ROPS). The compactor shall be equipped with a collapsible or removable bolted down ROPS. The ROPS shall conform to ISO 3471. A slinging eye(s) conforming to MIL-STD-209 shall be provided to facilitate ROPS removal.

3.5.3 Human factors engineering (HFE). The compactor shall be operable and maintainable by a small human through a large human as defined by ISO 3411 while wearing cold weather clothing

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

and nuclear, biological, chemical (NBC) protective gear. Controls and displays shall be located in accordance with ISO 6682 and SAE J297.

3.5.4 Noise limits. The noise produced by the compactor shall be measured in accordance with 4.5.2.6 of this specification and paragraph 5.1.2.3.2 of MIL-STD-1474 and conform with SAE J88. The noise level shall not exceed the steady state noise of category C (90 dB(A)). The noise level shall also not exceed 88 dB(A) when tested in accordance with SAE J88. If either sound level exceeds 85 dB(A), the appropriate provisions of paragraph 4.2 and 4.3 of MIL-STD-1474 shall be provided.

3.6 Steering system. The compactor shall be equipped with a hydraulic powered articulated (center point) type steering system which provides for tracking of front and rear wheels of full width coverage. Compactors, built in two sections with a hinge point for articulation, shall have steering frame locks conforming to ISO 10570. A caution plate shall be posted on each side of the compactor near the articulation point cautioning personnel not to stand in the area of the articulation point when the engine is running and to use the steering locks when servicing the equipment and during transport.

3.7 Lighting system. The compactor shall be furnished with front and rear mounted head lamps or flood lamps for illumination of the area in front of and behind the compactor for night operation of the compactor.

3.8 Propulsion system. The compactor shall be equipped with a torque converter and a power shift transmission having at least 3 gear speed ratios in both forward and reverse. A single forward-reverse lever shall be provided to control the direction of the compactor for each of the 3 minimum gear speed ratios. Engine power may also be transmitted through a full hydrostatic drive with infinitely variable speed control in forward and reverse. The compactor engine starting system shall be such that the transmission must be placed in neutral position before the engine can be started.

3.9 Brake system. The compactor shall be provided with a service and parking brake system which conforms to SAE J1472.

3.9.1 Brake fluid. When hydraulic brakes are utilized, the brake system shall be compatible with silicone brake fluid conforming to MIL-PRF-46176.

3.10 Strike-off (leveling) dozer blade. The leveling blade shall be hydraulically operated and shall have a replaceable cutting edge. The blade width shall be wider than the compactor width, but shall not exceed the compactor width by more than 6 inches on each side. The blade height shall be not less than 30 inches. Hydraulic controls shall include lift, lower, float and hold positions.

3.11 Wheels and cleaners. Each compactor wheel or drum shall be provided with tamping feet or segmented impact pads. The tamping foot tips or segmented impact pad caps shall be of the replaceable type. Adjustable and replaceable cleaners shall be provided fore and aft.

3.12 Control station. The compactor shall be equipped with a operator control station. The station shall be sized in accordance with ISO 3411. Access and egress from the station shall be in accordance with SAE J185.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

3.12.1 Seating. The control station shall be equipped with a seat conforming to SAE J899. The seat shall be covered with a durable, water resistant covering.

3.12.2 Instruments. The control station shall be equipped with all instruments, gauges, and indicators necessary for operation of the compactor. As a minimum, the control station shall be equipped with indicators which monitor amps or volts, oil pressure, fuel level, and engine coolant temperature. Pictograms conforming to SAE J1362 may be used in lieu of text labels.

3.12.3 Controls. The control station shall be equipped with all controls necessary for operation of the compactor. Controls shall be designed and located in accordance with SAE J297 and ISO 6682. Pictograms conforming to SAE J1362 may be used in lieu of text labels.

3.13 Diesel engine. The compactor shall be powered by a diesel engine having horsepower, torque, and speed characteristics to satisfactorily meet all performance requirements specified herein without exceeding the engine manufacturer's intermittent rated horsepower of the engine furnished. The engine governed speed shall be set and sealed in accordance with the engine manufacturer's recommendations. The engine shall operate on either JP-8 turbine or diesel fuel conforming to MIL-DTL-83133 or A-A-52557, respectively and shall meet all performance requirements specified herein when operating on either fuel.

3.13.1 Engine fuel tank. The fuel tank shall be corrosion resistant and of sufficient capacity for not less than 10 hours operation at the maximum fuel rating of the engine. The fuel tank shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Fuel lines shall be protected against mechanical damage. Construction and mounting shall be such that neither vibration nor tightening of mounted devices will cause leaks to develop. The bottom of the fuel tank and its supporting members shall not be lower than the lowest adjacent part of the chassis or body. A fuel line shutoff to the tank shall be provided if the fuel filter is mounted lower than the top of the fuel tank. The fuel fill port shall be labeled "DIESEL OR JP-8 FUEL ONLY" in letters not less than 1 inch height.

3.13.2 Oil sampling valve. An oil sampling valve conforming to MIL-V-81940/1B and SAE J514 shall be provided on the engine. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the valve shall be such that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. An oil sampling valve shall be provided for the transmission and hydraulic system. The sampling valve shall be labeled adjacent to the valve indicating the type of oil sampled (engine or transmission or hydraulic, as applicable).

3.13.3 Diagnostic connector assembly (DCA). When specified (see 6.2), the compactor shall incorporate an easily accessible DCA located in or near the operator's compartment for interface with the Simplified Test Equipment for Internal Combustion Engines Reprogrammable (STE/ICE) test equipment. The DCA shall be in accordance with TACOM drawing no. 12258941 and shall be labeled "Diagnostic Connector Assembly". All requirements for DCA shall be in accordance with STE/ICE-R design guide for vehicle Diagnostic Connector Assemblies, report no. CR-82-588-003, revision 1. The DCA shall have the capabilities for measuring the minimum functions defined in

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

table 2-2 of report no. CR-82-588-003, revision 1. Additional test parameters to be monitored in either the DCA or transducer kit (TK) mode shall be as follows: fuel return pressure, engine oil pressure, engine oil filter pressure drop, starter solenoid voltage, and transmission oil pressure. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Those additional test parameters which are inaccessible for monitoring without the removal of other components in the TK mode shall be monitored in the DCA mode. Fittings/adapters required for interface with the STE/ICE-R test set when monitoring in the TK mode, shall be permanently installed on the compactor to insure that adapters in the STE/ICE-R test set readily mate with the TK mode test point locations. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. A fuel shut-off method shall be provided for conducting compression unbalance checks. The contractor shall also provide STE/ICE-R vehicle test cards in the format specified in TACOM drawing 12258955 addressing both DCA and TK measurements, and shall incorporate the test cards into the compactor technical manuals.

3.14 Electrical system. The compactor shall be furnished with a 24 volt electrical system including a NATO slave receptacle conforming to MIL-PRF-62122. The connector shall be provided and installed on the exterior of the vehicle near the battery enclosure and shall be accessible to personnel on the ground. A data plate shall be provided adjacent to the slave receptacle which reads "24 volts".

3.14.1 Batteries. Batteries conforming to MS52149, type 6TL, shall be furnished and shall be accessible for service without removal of other components. At the contractor's option, commercial batteries with equal or greater capacity than the type 6TL military may be initially furnished, provided they are compatible with the size of the battery box and battery cables/ terminals such that military 6TL batteries will be interchangeable, as replacements and the box and terminals are sized to accommodate the quantity of 6TL batteries necessary to achieve performance of all requirements specified herein.

3.15 Electromagnetic interference. The Scraper shall comply with the Electromagnetic Radiation Operational requirements in accordance with MIL-STD-464, using TABLE 1C.

3.16 Bracket assembly, rifle stowage. A rifle stowage bracket assembly conforming to Drawing 13229E8014 shall be furnished and mounted by the contractor. The location of the bracket shall be such that it is accessible to the operator and provides clearance for the rifle to be removed. The bracket shall be located away from operator's controls and shall not impair the vision of the operator.

3.17 Transportability. The compactor shall be capable of being transported (see 6.2), by military or commercial trailers, rail, marine vessels, and aircraft and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The compactor shall be equipped with tiedown and slinging provisions conforming to MIL-STD-209. Optional provisions permitted by paragraph 4.1.3 of MIL-STD-209 are acceptable for the compactor and any components removed for transport. During preparation for highway or rail transport, the ROPS, blade, and other minor components may be removed provided disassembly and reassembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

can be accomplished in one man hour or less by two people using only hand tools and a lifting device and operator. Any components (except for exhaust stack) removed for transport must have tiedowns and slinging provisions conforming to MIL-STD-209. The compactor shall have 25 percent of its fuel supply and shall be fully serviced when transported.

3.17.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type III equipment. MIL-HDBK-1791 may be used for guidance on equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified in 4.5.2.8, without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Tiedown points shall be labelled, in minimum 1-inch high letters, "TIEDOWN HERE" and combination lift/tiedown points "LIFT/TIEDOWN HERE".

3.17.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type III. The provisions shall enable the complete compactor to be lifted in the normal operating position without the use of a spreader bar. The provisions shall be located so that not less than 1-inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.17.1. Lift points shall be labelled in minimum 1-inch high letters, "LIFT HERE".

3.17.3 Air transportability. The compactor shall be capable of being driven onto a C-5 aircraft without modification. See MIL-HDBK-1791 for guidance (see 6.7).

3.17.4 Rail transportability. The Compactor shall be rail transportable in the Continental United States (CONUS) and North Atlantic Organization (NATO) countries. The SSL must be capable of withstanding shock loads resulting from the rail impact testing specified in 4.5.2.10 without failure, damage, or permanent deformation. When loaded on a 50 inch high rail car, the Compactor shall meet the DOD clearance profile for the Strategic Rail Corridor Network (STRACNET). Transport without disassembly is desired.

3.17.5 Highway transportability. The compactor, when loaded on a semitrailer/tractor, shall be within the highway permit limits of all states.

3.17.6 Marine transportability. The compactor shall be transportable by LARC-V and larger vessels operating over any type of waterway. The compactor shall withstand, without damage, the shock, rolling, and pitching normally experienced in marine transportation on the deck or in the hold of the cargo vessel.

3.18 Towing attachments. Towing attachment(s) shall be provided. The attachment(s) shall be hook(s) or loop(s) suitable for towing (or recovering) the compactor.

3.19 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. The compactor shall be serviced with military lubricants selected by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

MIL-HDBK-113 may be used for guidance. MIL-PRF-2104 shall be used in the engine, transmission and hydraulic systems and SAE J2360 shall be used in the axles. Run-in and testing shall be performed with these lubricants. A lubrication data plate conforming to SAE J753 shall be mounted on each compactor and shall identify all military lubricants and applicable temperature ranges to be used when servicing the compactor. Grease lubrication fittings conforming to SAE J534 shall be provided in a protected, but accessible location.

3.19.1 Filling, draining, and checking provisions. Each enclosure that contains a reservoir of lubricant for the lubrication of parts enclosed shall be equipped with dipstick, check plug, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with lubricant and for draining. The drain outlet shall be located so that the removal of the plug will result in complete drainage of the lubricant. Drainage shall be to a container without contacting other components when the compactor is in its normal position.

3.19.2 Servicing and adjustments. The contractor shall service and adjust each compactor for correct operation and travel.

3.20 Data plates. All new plates shall conform to A-A-50271, contractors recommended size, Composition C, with black background and silver lettering (except in those instances where black background conflicts with prescribed colors of warning or caution data plates). All data plates shall be attached by screws, bolts, or rivets, a conspicuous protected location unless otherwise approved by the Contracting Officer.

3.20.1 Shipping data plate. A shipping data plate shall be furnished showing the silhouette of the compactor in transport position and indicate the center of gravity, weight of the compactor, and the location and capacity of the lifting and the tiedown provisions. The recommended lifting and tiedown arrangement shall be included on the plate.

3.20.2 Identification marking. Each compactor shall be furnished with an identification plate showing the contractor's model number, NSN, USA registration number, and date of manufacture, contractor's name, and contract number. The plate shall be affixed to the compactor in a conspicuous location. ROPS and blade, if removed or disassembled for transport, shall be match marked for proper reassembly. Registration numbers will be furnished by the contracting officer and stenciled on the vehicles.

3.21 Treatment and painting. The portions of the compactor normally painted, internally and externally, shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. The top coat color shall be 34094 green 383 conforming to FED-STD-595 with the exception that engine, transmission, and axles which are not visible during normal operation may be painted with the manufacturer's commercial paint and color. When specified (see 6.2), the compactor shall be over-coated in accordance with the Government furnished camouflage patterns. High temperature surfaces shall not be painted with CARC paint but rather a high temperature paint in a color consistent with specified vehicle paint requirements, i.e. green 383 or camouflage pattern scheme.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

3.21.1 Camouflage pattern data. When specified (see 6.2), the contractor shall provide drawings for development of camouflage pattern data by the government.

3.22 Tool box. The compactor shall be provided with a lockable, fully enclosed and water resistant compartment or tool box to carry tools and accessories required for operation and operator maintenance of the equipment.

3.23 Workmanship. Each compactor shall have no evidence of cracks, dents, scratches, burrs, sharp edges, loose parts, missing fasteners, distortions, spatter, or foreign material. Workmanship of the compactor shall in no case, be less than that of comparable products furnished to the commercial market.

3.24 Government-furnished property. When specified (see 6.2), the following property will be furnished or loaned by the Government (see 6.8).

Item No.	Description
1	Camouflage paint pattern.
2	Report CR 82-588-003 STE/ICE Design Guide for Vehicle Diagnostic Connector.
3	STE/ICE-R-test set.

NOTE: DCA tester identified in paragraph 4.5.2.12 is the contractor's responsibility.

3.25 Reliability. When specified (see 6.2), a reliability test shall be conducted by the government. The specified mean time between hardware failure (MTBHF) shall be 75 hours when tested in accordance with 4.5.2.15. For the purpose of this test, a failure is defined as any malfunction which precludes the compactor from performing mobility and compaction in its intended mission, as described in 6.1, and could not be corrected within 30 minutes by the operator using only on-board tools.

4. VERIFICATION.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.2 Reserved.

4.3 First article inspection.

4.3.1 Examination. The compactor shall be examined as specified in 4.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

4.3.2 Test. The compactor shall be tested as specified in 4.5.2.1 through 4.5.2.15. All first article tests shall be run using JP-8 fuel conforming to MIL-DTL-83133.

4.4 Quality conformance inspection.

4.4.1 Examination. Each compactor shall be examined as specified in 4.5.1.

4.4.2 Test. Each compactor shall be completely assembled, properly adjusted, and otherwise serviced for operation. Each compactor shall be run in and in accordance with the manufacturer's standard commercial production run in procedures. All controls shall be operated as many times as necessary to determine ease of operation, effectiveness, and responsiveness, and that the mechanisms actuated by the controls operate promptly, fully, and without restriction, malfunction, or excessive vibration. Gear cases, bearings, and other moving parts shall be examined for excessive heating or abnormal operation insofar as is practicable and possible. Tests shall be run using JP-8 or diesel fuel conforming to MIL-DTL-83133 and A-A-52557, respectively (see 3.13). The entire compactor shall be examined for evidence of failure, slippage, or deformation.

4.5 Inspection procedures.

4.5.1 Examination. The compactor shall be examined for the following defects:

101. Compactor not assembled as specified (see 3.1).
102. Dimensions not as specified (see 3.1.2).
103. Materials not as specified (see 3.4).
104. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.4.1).
105. Dissimilar metals are not effectively insulated from each other (see 3.4.2).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the compactor (see 3.4.3).
107. Radioactive materials used (see 3.4.4).
108. Asbestos materials used (see 3.4.5).
109. Maintainability not as specified (see 3.5.1).
110. Safety devices missing or not as specified (see 3.5.2).
111. Rollover protective structure (ROPS) not as specified (see 3.5.2.1).
112. Human factors engineering requirements not as specified (see 3.5.3).
113. Steering system not as specified (see 3.6).
114. Lighting system not as specified (see 3.7).
115. Propulsion system not as specified (see 3.8).
116. Brake system not as specified (see 3.9).
117. Brake fluid not as specified (see 3.9.1).
118. Strike-off blade not as specified (see 3.10).
119. Wheels and cleaners not as specified (see 3.11).
120. Control station not as specified (see 3.12 thru 3.12.3).
121. Engine not as specified (see 3.13).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

122. Engine fuel tank not as specified (see 3.13.1).
123. Oil sampling valve(s) not as specified (see 3.13.2).
124. Diagnostic connector not as specified (see 3.13.3).
125. Electrical system not as specified (see 3.14).
126. Batteries not as specified (see 3.14.1).
127. Rifle stowage bracket assembly not as specified (see 3.16).
128. Transportability not as specified (see 3.17 thru 3.17.6).
129. Towing attachments not as specified (see 3.18).
130. Lubrication not as specified (see 3.19).
131. Filling, draining, and checking provisions not as specified (see 3.19.1).
132. Servicing and adjustments not as specified (see 3.19.2).
133. Data plates missing or not as specified (see 3.20).
134. Shipping data plate not as specified (see 3.20.1).
135. Identification marking not as specified (see 3.20.2).
136. Treatment and painting not as specified (see 3.21).
137. Tool box not as specified (see 3.22).
138. Workmanship not as specified (see 3.23).

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed at the climatic conditions existing at the place of test. The compactor shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.5.2.2 Performance. Operate the compactor on a fill, built over a natural soil subgrade having an initial density of at least 90 percent when measured in accordance with ASTM D1557. The fill material shall correspond to two different groups of soils: a clayey sandy (sc) type with a plasticity index of 5-12, and a lean clay (cl) type with a plasticity index of 10-16, of the Soil Classification Chart in ASTM D2487. Both types of soils, when compacted shall be within 2 percentage points of their respective optimum moisture content according to ASTM D1557, methods B, C, or D. The compacted fill site shall be not less than 30 feet wide, with the actual width of the fill being a minimum of 3 feet wider than the compacted area as shown in figure 1. The fill may be placed prior to the start of compaction operation, during compaction operations, or the contractor may elect to till the natural soil in lieu of placing fill material. When hauling equipment is run over spread, un-compacted fill material, the fill site shall be scarified or tilled to a uniformly loose un-compacted condition. Spreading of the fill material shall be done with a grader. Spreading runs shall be only that necessary to reach the estimated un-compacted fill lift thickness. Water distribution equipment may be used as required. The material to be compacted shall be of sufficient thickness, initially, to give a compacted lift of at least 6-inches. The laboratory density of the soil to be compacted shall be determined in accordance with ASTM D1557, method optional. Compaction operations may be stopped by the contractor only after the following conditions have been met: Exactly 1 continuous hour has elapsed from the time the test has begun to determine whether or not the compactor is successfully able to compact the type(s) of soils described herein to 1500 cubic yards per hour with the minimum lift and test site dimensions/restrictions cited herein; and, the soils tested have been

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

compacted to 90 percent of maximum density at optimum water content according to ASTM D1557. The time used in density testing shall not be counted as part of the ten (10) hour operating time specified herein. Density measurements shall be taken at points along the test strip using the pattern shown on figure 1 at intervals of 200 feet. In addition, a minimum of three (3) samples shall be taken at intervals of 100 feet and at random locations specified by the Government. Nuclear density testing shall be accomplished according to ASTM D6938, direct transmission method. The depth of measurement shall correspond to the compacted soil layer thickness (minus 1 inch maximum). Oven dried laboratory soil moisture content tests shall be used to check, and correct if necessary, the nuclear moisture reading. An achieved full depth fill material density of less than 95 percent of maximum density at optimum water content according to ASTM D1557, or compacted fill layer thickness of less than 6 inches in depth shall constitute failure of this test. Sufficient measurements shall be taken to determine the in place quantity of material compacted. The above procedure shall be repeated for a total of 3 times for each type of soil. The average compacted cubic yards shall be determined. After all compaction measurements have been computed, the compactor shall continue to compact soils at the 1500 cubic yard per hour rate to verify conformance.

4.5.2.2.1 Failure criteria. An average compaction rate of less than the compacted cubic yards per hour specified in 3.1.3 and failure to achieve full depth material density or minimum compacted fill layer thickness as specified in 4.5.2.2 for each test shall constitute failure of this test.

4.5.2.3 Gradeability. The compactor shall be operated to demonstrate the capability of light bulldozing, and leveling of loose material, the ability to climb a 30 degree slope, and turning and maneuvering on a 15 degree slope as specified in 3.3.

4.5.2.3.1 Failure criteria. Inability to negotiate the grade under either condition specified above shall constitute failure of this test.

4.5.2.4 High and low temperature operation. The contractor shall provide proof (test data or calculation) that the compactor can be operated in ambient temperatures from -25°F to +125°F without exceeding the engine manufacturer's recommended limits for radiator top tank temperature, engine oil temperature in the sump and gallery, and hydraulic component manufacturer's fluid temperature limit.

4.5.2.4.1 Failure criteria. Failure to meet this requirement shall constitute failure of this test.

4.5.2.5 Electromagnetic interference. Unless otherwise specified, the compactor shall be tested in accordance with MIL-STD-461 or a certification provided by the contractor confirming that the compactor conforms to paragraph 3.15. Certification shall contain technical analysis which forms the basis of the determination that the compactor conforms to MIL-STD-461, and thereby justifies not conducting the test.

4.5.2.5.1 Failure criteria. Failure to meet the requirements of MIL-STD-461, shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

4.5.2.6 Noise level test. Noise levels shall be measured in accordance with paragraph 5.1.2 of MIL-STD-1474 and reported in the format specified by paragraph 5.5 of MIL-STD-1474 requirements specified herein. Noise contours shall be taken at not fewer than 12 equal horizontal arc increments, with data recorded at the noisiest position. The noise levels at typical operating position shall also be measured and recorded.

4.5.2.6.1 Failure criteria. Failure to comply with MIL-STD-1474 requirements specified herein or 3.5.4 provisions shall constitute failure of this test. Noise levels shall also be recorded when tested in accordance with SAE J88. For purposes of defining duty cycle, the noise test shall be conducted during testing specified in paragraph 4.5.2.2

4.5.2.7 Brake test. The compactor shall be completely stopped five times within a period of 10 minutes, to test brake fade out. The distance from point of brake application to actual stop shall be measured on the fifth pass to determine conformance to 3.9. The service brake test shall be conducted in both forward and reverse. With the compactor stopped on a 15 degree slope, the parking brakes shall be applied and shall be capable of securing the compactor in the stopped position without creeping. The brake test shall be conducted with the compactor heading up and down slope.

4.5.2.7.1 Failure criteria. Failure of the compactor to meet the braking requirements specified in 3.9 shall constitute failure of this test.

4.5.2.8 Tiedown provision test. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.17.1. Inability to meet the requirements of 3.17.1 shall constitute failure of this test.

4.5.2.9 Slinging provision test. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.17.2. Inability to meet the requirements of 3.17.2 shall constitute failure of this test.

4.5.2.10 Rail transportability test. Rail impact shall be tested in accordance with MIL-STD-810 to prove conformance to 3.17.4. Inability to meet the requirements of 3.17.4 shall constitute failure of this test.

4.5.2.11 Air transportability test. When required by the Government, the contractor shall provide one compactor for air transportability testing. The air transportability shall be tested in accordance with MIL-STD-1792 to prove conformance to 3.17.3 at a Government test site. Inability to meet the requirements of 3.17.3 shall constitute failure of this test.

4.5.2.12 DCA capability. The contractor shall perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the compactor to ensure wire and connections are free of error. After installation on the compactor, test the DCA harness (see Report No. CR-82-588-003) on each vehicle with a DCA tester prior to interfacing with STE/ICE.

4.5.2.12.1 Failure criteria. Non-conformance to 3.13.3 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

4.5.2.12.2 DCA-STE/ICE interface. Upon completion of inspection and tests specified in 4.5.2.12, test the DCA for all measurements specified in 3.13.3 using the STE/ICE.

4.5.2.12.2.1 Failure criteria. Non-conformance to 3.13.3 or failure of any component or wiring in the DCA harness which results in inability to obtain the measurements previously recorded with the STE/ICE shall constitute failure of this test.

4.5.2.13 Roll over protection structure (ROPS). The contractor shall provide a certified test report of prior testing, or test the ROPS, to demonstrate conformance to ISO 3471. Insufficient test data in the report or failure of the ROPS during the test shall constitute failure of this requirement.

4.5.2.14 Low temperature. The compactor shall be cold soaked for 24 hours at -25°F, and then started. Failure to start and operate as specified herein, or non-conformance to paragraph 3.3 shall constitute failure of this test.

4.5.2.15 Reliability. If a reliability test is specified (see 6.2), the Government will select a quantity of compactors to be subjected to a Government conducted test consisting of operations in accordance with the compactor's intended mission (see 6.1). Quantity of compactors and duration of test will be that which is necessary to confirm the production compactors demonstrate the reliability requirement of 3.25. Compactors not achieving the reliability requirement shall constitute failure of the test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The high speed, segmented compactor covered by this specification will be used for embankment compaction of SC type soils during the construction of roads, airfields, and dams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. When a first article is not required (see 3.2).
d. Time frame for submission of first article and number of compactors required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests (see 3.2). When the Government will conduct some but not all of the first article examination and tests the contracting officer should specify which examination and test will be conducted by the Government and which examination and test will be conducted by the contractor (see 3.2).
f. When electromagnetic interference requirements are other than as specified (see 3.15).
g. When over-coating with camouflage patterns is required (see 3.24).
h. When camouflage pattern drawings and camouflage patterns will be provided (see 3.21.1 and 3.24).
i. Degree of preservation and packing required (see 5.1).
j. When a transportability report is required (see 6.7).
k. When DCA is required (see 3.13.3).
l. When reliability testing is required (see 3.25).

6.3 First article. When a first article inspection is required, the item(s) should be a pre-production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Commerciality. When requested by the procuring activity, the contracting officer should require the contractor to submit evidence to establish commerciality of the compactor and furnish names and addresses of non-Government sources that have acquired the compactor at least one year prior to the opening date of the solicitation.

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of compactors.

6.7 Transportability report. When required (see 6.2), the contracting officer should require the contractor to provide sufficient information (dimensions and weight) to the Government to confirm that the compactor being supplied can be loaded and transported by C-5 aircraft.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.24.

6.9 Subject term (key word) listing.

Diesel engine
Embankment
High speed roller
Landfill roller
Steel wheel roller
Tamping foot

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

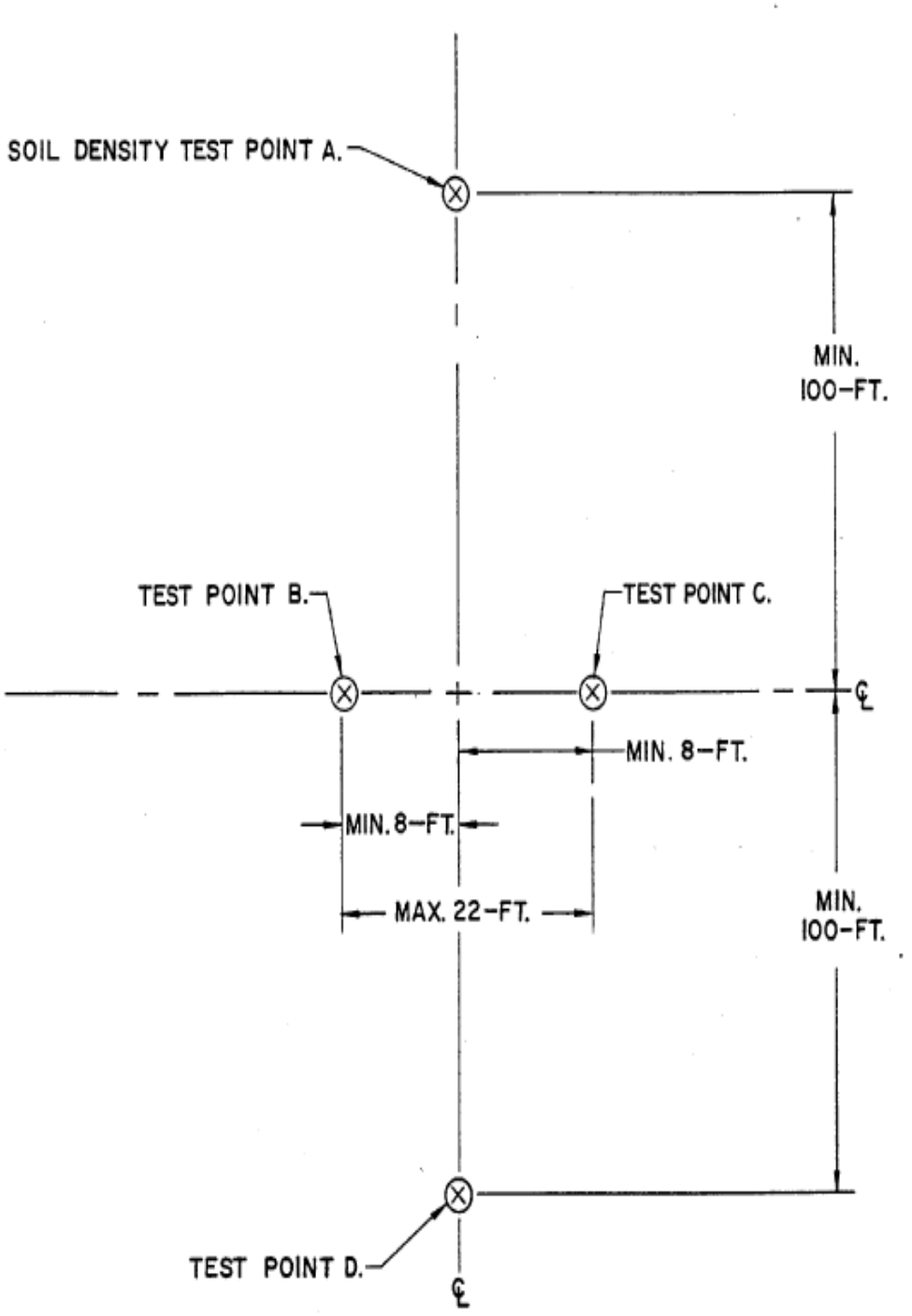


Figure 1 Test strip pattern.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project 3895-2012-001

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:50Z
Check the source to verify that this is the current version before use.